

File S1. Pre-specified study design and log of deviations

Pre-specified research design (written before any model was fitted)

Date fixed: 2026-10-03. Git commit time-stamps this document before any experiment is run. Every later deviation is listed in `supplementary/S9_deviations.md`.

1. Problem

Breeders use genomic predictions to choose which candidates to keep. A useful prediction needs an honest statement of uncertainty, and the reliability of a prediction depends strongly on how closely the candidate is related to the training (reference) population. This is classical quantitative genetics: prediction error variance (PEV), reliability, and Habier/Clark/Pszczola-type relatedness effects.

Current uncertainty statements come in four kinds: * model-based Gaussian intervals from the PEV of the mixed model; * Bayesian posterior predictive intervals; * recently, split-conformal intervals (CalGS, a preprint on simulated data); * in human polygenic scores, context-calibrated intervals (CalPred, Hou et al. 2024) and cross-validation residual quantiles (PredInterval).

Split and CV conformal guarantees assume exchangeability between calibration and deployment individuals. In breeding, deployment candidates often come from new families or sub-populations. Their relatedness to the training set therefore differs systematically from that of a random calibration hold-out, which breaks exchangeability. Model-based intervals do adapt to relatedness through PEV, but they rely on correct model assumptions: Gaussianity, additivity, and correctly estimated variance components.

2. Hypothesis

Relatedness to the training set, summarised for each candidate by its GBLUP prediction error variance in σ^2_g units ($\mathbf{d}$), is the main axis along which genomic-prediction uncertainty statements become miscalibrated.

A conformal procedure that does three things will give near-nominal coverage that is both marginal and relatedness-conditional, across deployment regimes and species, at modest cost: * (A) builds a calibration pool covering the relevant relatedness range, using residuals from both random-fold and cluster-fold cross-fitting inside the training set; * (B) normalises nonconformity scores by the PEV-implied predictive standard deviation; * (C) localises the conformal quantile in $\mathbf{d}$ (kernel weighting).

The proposed method is called **KinCP** (kinship-aware conformal prediction).

3. Research questions → experiments

RQ	Question	Experiments
RQ1	How miscalibrated are standard genomic-prediction intervals across relatedness levels and deployment regimes?	E1, E2
RQ2	Does KinCP restore marginal and relatedness-conditional coverage, and which components are responsible?	E1, E2, E3
RQ3	Does this hold across species, traits, base predictors, training-set sizes and simulated genetic architectures with known truth?	E4, E5
RQ4	What does it cost computationally, and how sensitive is it to its tuning choices?	E6, E7
RQ5	Where does it fail, and does calibrated uncertainty change selection decisions?	E8, E9

4. Data

EasyGeSe (Quesada-Traver et al. 2025, BMC Genomics; Zenodo 10.5281/zenodo.15348871; CC-BY 4.0). Ten species: barley, common bean, lentil, loblolly pine, eastern oyster, maize, pig, rice, soybean, wheat. Genotypes and phenotypes are used exactly as distributed.

Trait selection rule (fixed in advance; does not look at results). A trait is eligible if it has ≥ 200 non-missing records and ≥ 30 distinct values, i.e. it is quasi-continuous. Within each dataset we take up to **3** eligible traits, chosen by largest non-missing n. Ties are broken by column order in the distributed file.

5. Deployment regimes

- **R1 Random:** the five EasyGeSe "Split1–5 $\times$ CV" 5-fold assignments, i.e. 5 repeats $\times$ 5 folds.
- **R2 Cluster-out:** leave-one-genomic-cluster-out. Clusters come from k-means ($k = 5$, seed 2026, 10 restarts) on the top 10 principal components of the standardised genotype matrix. This uses genotypes only and never looks at phenotypes.
- **R3 Reduced training:** R1 folds with the training set subsampled to 50% (seeded).

6. Base predictors

- GBLUP: VanRaden GRM, REML via eigendecomposition. This is the primary predictor.
- RKHS: Gaussian-kernel ridge, median-heuristic bandwidth, λ chosen by internal GCV.
- LightGBM: fixed default-style hyper-parameters, decided in advance (500 trees, learning rate 0.05, 31 leaves, colsample 0.3). No tuning on test folds.

7. Uncertainty methods (nominal $1-\alpha = 0.90$; $\alpha \in \{0.05, 0.10, 0.20\}$ for sensitivity)

1. **Gauss-PEV:** GBLUP model-based interval $\hat{y} \pm z \cdot \sqrt{\sigma^2_g \cdot d + \sigma^2_e}$. GBLUP only.
2. **Gauss-homosc:** $\hat{y} \pm z \cdot \text{SD}$ of out-of-fold training residuals.
3. **SCP:** split conformal, with a random 20% calibration hold-out from the training set.

4. **CV+**: cross-conformal / CV+ (Barber et al. 2021), $K = 5$ random folds.
5. **CalPred-style**: heteroscedastic Gaussian with $\log \sigma$ linear in $\mathbf{d}$, fitted by maximum likelihood to out-of-fold residuals.
6. **NormCP**: split conformal with scores normalised by $\sqrt{\sigma^2_g \cdot \mathbf{d} + \sigma^2_e}$, random calibration. This is component B alone.
7. **KinCP (proposed)**: components A + B + C.

Ablations (E3): KinCP with one or two of A, B, C removed, giving all 7 non-empty sub-combinations plus none ($\approx$ CV+).

8. Endpoints (pre-specified)

- **Primary 1**: relatedness-conditional coverage error. Test individuals in each dataset $\times$ trait $\times$ regime are pooled over folds, binned into quintiles of $\mathbf{d}$, and we report the mean over quintiles of $|\text{coverage}_q - 0.90|$.
- **Primary 2**: marginal coverage deviation $|\text{coverage} - 0.90|$.
- **Primary 3**: mean interval score (Winkler) divided by the trait's phenotypic SD.
- **Secondary**: mean width divided by SD; worst-quintile coverage; runtime.

9. Statistics

- **Unit of analysis**: the dataset $\times$ trait combination, crossed with regime.
- **Paired comparisons** of KinCP against each competitor: two-sided Wilcoxon signed-rank test with Holm correction within each endpoint family. Effect sizes are the median paired difference with a 95% bootstrap CI (10,000 resamples) and the matched-pairs rank-biserial correlation.
- **Seeds**: stochastic components use seeds {11, 22, 33, 44, 55}, fixed in advance. Results are reported as mean $\pm$ SD over seeds, and the best seed is never selected.

10. Simulation with known truth (E5b)

- Real genotypes from pine, pig and maize.
- Additive traits simulated with $h^2 \in \{0.2, 0.5, 0.8\}$ and $n_{\text{QTL}} \in \{10, 1000\}$, with effects normally distributed.
- Five replicates.
- Phenotype coverage is evaluated together with diagnostics against the true genetic value.

11. Leakage controls

- Phenotype standardisation, REML variance components, hyper-parameters and calibration scores all use training individuals only.
- Building the GRM from all genotypes is allowed: candidate genotypes are known at selection time and no phenotypes are involved. This is stated explicitly.
- Duplicate or near-identical genotypes (standardised relationship > 0.95 between distinct individuals) are reported, and a sensitivity analysis removes them.

Deviations from the pre-specified design (S0), in chronological order

#	When	Deviation	Reason	Decided before seeing main results?
D1	Pipeline smoke test (oyster, one trait, one repeat)	Localisation component C was given a bandwidth floor. A test row's kernel bandwidth is widened ($h \leftarrow 1.25h$) until the total calibration weight $\sum w_i$ (in units of the test point's own weight, $w = 1$) is $\geq n_{\min} = 50$. A first variant based on the Kish effective sample size was replaced in the same smoke test, because uniformly tiny weights can have a large Kish ESS and still give an infinite quantile. The fixed-bandwidth version is kept and reported as the sensitivity variant <code>KinCP[fixed-h]</code> , together with $n_{\min} \in \{25, 100\}$.	With a fixed bandwidth, test individuals whose relatedness covariate lies outside the calibration pool's range get near-zero total weight. Their conformal quantile is then $+\infty$, i.e. an uninformative interval. The floor depends only on genotype-derived covariates, never on outcomes.	Yes. Only the smoke-test output was seen: one oyster trait, one repeat, used to check the code runs.
D2	Implementation	LightGBM uses at most 10,000 evenly spaced markers, and is run for R1 repeat 1 and R2 seed 11 only. RKHS is run for R1 repeat 1 and R2 seeds 11 and 22.	Compute budget (4 CPU cores, no GPU). Twelve model fits are needed per outer fold.	Yes
D3	Implementation	Inner cross-fitting reuses the variance ratio δ estimated by REML on the outer training fold, instead of re-estimating it in every inner fold.	Compute budget. It also keeps all variance components outer-fold-only.	Yes
D4	Implementation	Individuals that have a phenotype but no EasyGeSe fold label are assigned to folds deterministically (round-robin). How many this affects is logged per dataset in <code>results/data_audit.json</code> .	The EasyGeSe CV files may omit some individuals.	Yes
D5	During the full run	LightGBM is not run on the maize dataset ($n = 4,421$). The GBLUP and RKHS analyses of maize are unchanged.	One LightGBM fit takes about 40 s for $n \approx 1,100$ on one core. The 12 fits per outer fold would need more than 12 CPU-hours for maize.	Decided on compute grounds only. Partial GBLUP/RKHS results for the other datasets had already been viewed, but no LightGBM result had.
D6	After the main real-data results were seen (Phase 19)	An exploratory variant was added, KinCP-G. Its inner cluster folds are the global genomic clusters present in the training set, so the calibration shifts match the granularity of cluster-out deployment. It is run on GBLUP with R1 repeat 1 and R2 seed 11 (tag explore). It is reported separately as exploratory and does not replace the pre-specified KinCP.	Hypothesis: KinCP's residual under-coverage in the least-related quintile under R2 (worst-quintile coverage 0.84) arises because inner k-means clusters of the training set are finer, and so less distinct, than the held-out deployment clusters.	No. Motivated by the observed results, hence labelled exploratory.
D7	Internal review, rounds 3–6, after the main results were seen	Robustness analyses were added: (i) cluster-out with $k = 3$ and $k = 10$ deployment clusters (GBLUP, one seed, all units); (ii) leave-one-F2-family-out for eastern oyster; (iii) conditional coverage on quintiles of maximum genomic relationship; (iv) species-level paired tests; (v) PEV-calibration diagnostic in the simulation; (vi) a scaling benchmark of time and memory; (vii) oracle width-rescaling and group-CV+ competitors (Reviewer 8).	These were requested by the internal adversarial reviews to test external validity, the endpoint, the unit of analysis and cost.	No. They are reported as robustness and sensitivity analyses, and none replaces a pre-specified endpoint.